
Automatic Attendance System

UML Activity Diagram Report

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This report presents the UML Activity Diagrams designed for a prototype Automatic Attendance System. The system uses a **proprietary barcode ID** to mark employee attendance, display profile information on screen, and allow admins to manage employee records. Two diagrams are included, each with a brief explanation.

1 Employee Attendance Flow

This diagram covers two phases. The first phase shows the complete flow from the moment an employee scans their barcode to the attendance record being saved and their profile confirmed on screen — including the error path for invalid or external barcodes. The second phase, separated by a dashed horizontal line, shows the automated end-of-day absent detection triggered by the Scheduler.

Swim lanes: Employee | System | Database | Scheduler

Step	Description
Phase 1: Attendance	
Present barcode	Employee presents their proprietary barcode ID at the terminal.
Read & decode ID	System reads and decodes the scanned barcode.
Valid format?	System performs a proprietary check on the ID format and checksum. External barcodes (e.g. Google Lens) fail here.
No — Show error	An error message is shown (Invalid ID). The flow ends without logging attendance.
Yes — Query employee	System queries the database for the employee record.
Fetch employee data	Database returns the matching employee data.
Log attendance	System writes a timestamped entry (Status: Present) to the database.
Save attendance record	Database saves the record with Status: Present .
Display profile	System displays the employee's Name, ID, Domain, and Photo on screen.
View profile	Employee sees the confirmation — Attendance Confirmed.
Phase 2: Absent Detection	
Trigger	Scheduler fires the end-of-day scheduled job automatically.
Run end-of-day job	Scheduler initiates the absent detection process.
Fetch all IDs	Database returns all registered employee IDs for the current day.
Cross-check scan log	System checks who has not scanned during the day.
Unmarked employees?	Decision: if no unmarked employees, the flow ends.
Mark as absent	System marks unmarked employees with Status: Absent .
Save absent record	Database saves the absent record with Status: Absent .
Notify admin	System sends the absent list for the day to the Admin.
Admin reviews	Admin reviews the absent list.

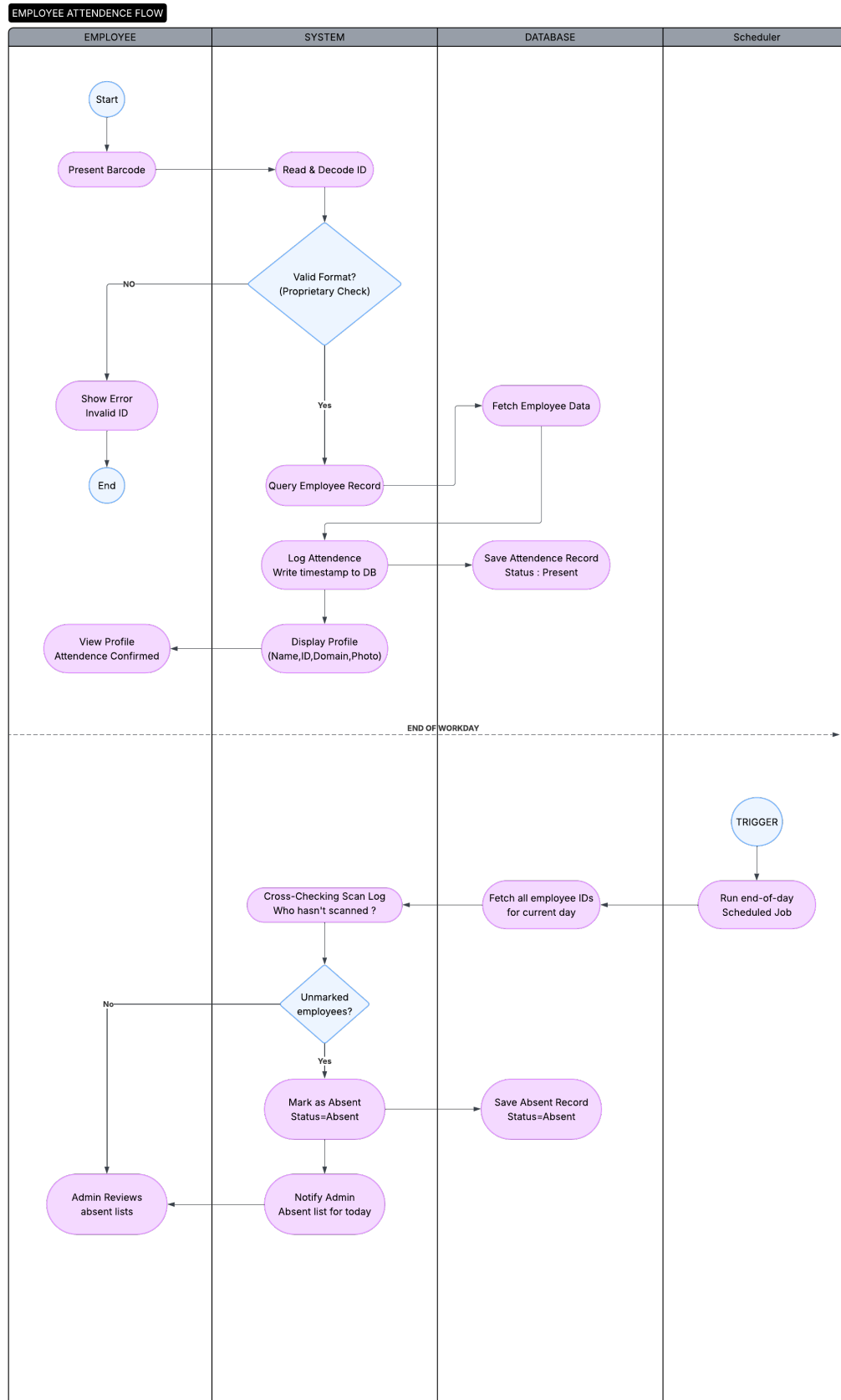


Figure 1. UML Activity Diagram — Employee Attendance Flow (including Absent Detection)

2 Admin CRUD Flow

This diagram describes how an administrator logs into the system and manages employee records through four operations — Create, Read, Update, and Delete. The diagram is split into three phases separated by dashed horizontal lines.

Swim lanes: Admin | System | Database

Phase / Step	Description
Phase 1: Login	
Open admin panel	Admin opens the panel and enters their credentials.
Verify credentials	System verifies the supplied credentials.
Validate	Decision: if credentials are invalid, a login error is shown and the Admin retries. On success, the flow continues.
Load dashboard	System loads the dashboard and displays the CRUD menu.
View dashboard	Admin views the dashboard and picks an operation.
Phase 2: Select CRUD Operation	
Choose operation	Admin selects one of four operations from the decision node.
Create	Admin enters employee details (Name, Domain, Contact). System generates a domain-specific ID (e.g. HPS250001) and saves the record and barcode to the database.
Read	System fetches attendance logs filtered by date or domain. Database returns the records. System displays or exports the report.
Update	Admin selects an employee and edits their profile fields. Database saves the updated record.
Delete	Admin confirms deletion via a confirmation pop-up. Database deletes the record and revokes the barcode ID.
Phase 3: Post-operation Confirmation	
Show success pop-up	System shows a success pop-up confirming the operation.
Acknowledge result	Admin acknowledges the result and chooses to continue or logout.

